Proiect Baze de Date

Magazin Virtual

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Grupa: C. 3.2

Universitatea Politehnica Timișoara

Aplicația reprezintă un sistem de gestiune pentru un magazin online, dezvoltat în mediul Oracle APEX pe baza unei structuri de 5 tabele relaționale *(Categorii, Producători, Produse, Comenzi și Linii Comandă)*.

Proiectul a fost realizat în echipă, efortul fiind împărțit în mod egal: **Studentul A** s-a ocupat de interfața Master-Detail pentru comenzi, rapoartele parametrizate pe categorii și centralizarea financiară a vânzărilor; **Studentul B** a implementat formularele de inserare pentru produse, interogările de căutare după CNP client și rapoartele statistice pe producători. Toate paginile integrează validări de date și clauze SQL de agregare pentru calculul automat al stocurilor, prețurilor medii și sumelor încasate.

Script ex1:

```sql
-- 1. Crearea tabelei mCategorie
CREATE TABLE mCategorie (
    tipCat VARCHAR2(5) PRIMARY KEY,
    categorie VARCHAR2(50) NOT NULL,
    tva NUMBER(4,2) NOT NULL
);

-- 2. Crearea tabelei mProducator
CREATE TABLE mProducator (
    codPr VARCHAR2(5) PRIMARY KEY,
    nume VARCHAR2(50) NOT NULL,
    adresa VARCHAR2(100),
    tel VARCHAR2(15)
);

-- 3. Crearea tabelei mProdus
CREATE TABLE mProdus (
    codP VARCHAR2(20) PRIMARY KEY,
    tipCat VARCHAR2(5) NOT NULL,
    denumire VARCHAR2(100) NOT NULL,
    codPr VARCHAR2(5) NOT NULL,
    stoc NUMBER(5) DEFAULT 0,
    pret NUMBER(10,2) NOT NULL,
    CONSTRAINT fk_produs_cat FOREIGN KEY (tipCat) REFERENCES mCategorie(tipCat),
    CONSTRAINT fk_produs_prod FOREIGN KEY (codPr) REFERENCES mProducator(codPr)
);

-- 4. Crearea tabelei mComanda
CREATE TABLE mComanda (
    cID VARCHAR2(20) PRIMARY KEY,
    cnp VARCHAR2(13) NOT NULL,
    nume VARCHAR2(50) NOT NULL,
    data DATE NOT NULL
);

-- 5. Crearea tabelei mListaCom
CREATE TABLE mListaCom (
    cID VARCHAR2(20) NOT NULL,
    poz NUMBER(3) NOT NULL,
    codP VARCHAR2(20) NOT NULL,
    buc NUMBER(4) NOT NULL,
    CONSTRAINT pk_mListaCom PRIMARY KEY (cID, poz),
    CONSTRAINT fk_lista_comanda FOREIGN KEY (cID) REFERENCES mComanda(cID),
    CONSTRAINT fk_lista_produs FOREIGN KEY (codP) REFERENCES mProdus(codP)
);
```

Script ex2:

```sql
INSERT INTO mCategorie (tipCat, categorie, tva) VALUES ('pi', 'PC-uri si IT', 21);
INSERT INTO mCategorie (tipCat, categorie, tva) VALUES ('ta', 'Televizoare si Audio', 19);
INSERT INTO mCategorie (tipCat, categorie, tva) VALUES ('tm', 'Telefoane Mobile', 22);


INSERT INTO mProducator (codPr, nume, adresa, tel) VALUES ('ap', 'Apple', 'Calea Floreasca nr. 169, Bucuresti', '0754122465');
INSERT INTO mProducator (codPr, nume, adresa, tel) VALUES ('sg', 'Samsung', 'Bd. Iuliu Maniu nr. 7, Bucuresti', '0770626214');
INSERT INTO mProducator (codPr, nume, adresa, tel) VALUES ('ln', 'Lenovo', 'Str. Aristide Demetriade nr. 1, Timisoara', '0733998877');
INSERT INTO mProducator (codPr, nume, adresa, tel) VALUES ('sn', 'Sony', 'Bd. Republicii nr. 15, Cluj-Napoca', '0800872678');


INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('AP-MQ24', 'pi', 'PC iMac 24"', 'ap', 23, 5999.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('AP-IP14P', 'tm', 'iPhone 14 Pro Max', 'ap', 31, 4200.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('AP-IP17P', 'tm', 'iPhone 17 Pro', 'ap', 12, 6500.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SG-LC27G5', 'pi', 'Monitor Gaming Samsung Odyssey G5', 'sg', 7, 2500.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SG-HG50U', 'ta', 'Televizor Smart Samsung Hotel UltraHD', 'sg', 21, 3011.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SG-S928B', 'tm', 'Samsung Galaxy S24 Ultra', 'sg', 16, 5429.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('LN-20XW', 'pi', 'Laptop Lenovo ThinkPad X1 Carbon', 'ln', 12, 6500.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('LN-L790', 'tm', 'Telefon Lenovo Legion Phone Duel', 'ln', 14, 3500.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('LN-55U8', 'ta', 'Televizor Smart LED Lenovo 139cm', 'ln', 5, 2100.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SN-OLED65', 'ta', 'Televizor Smart OLED Sony Bravia', 'sn', 8, 6500.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SN-WH5', 'ta', 'Casti Wireless Sony WH-1000XM5', 'sn', 20, 1550.00);
INSERT INTO mProdus (codP, tipCat, denumire, codPr, stoc, pret) VALUES ('SN-SB400', 'ta', 'Soundbar Sony cu Subwoofer Wireless', 'sn', 14, 1100.00);


INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034260', '1900515351234', 'Popescu Mathias', TO_DATE('16.04.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034261', '1851120410456', 'Stoia Marin', TO_DATE('22.04.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034262', '5010310127890', 'Roman Cristian', TO_DATE('27.04.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034263', '1950822243219', 'Ioan Marian', TO_DATE('01.05.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034264', '2920418354561', 'Toma Mariana', TO_DATE('03.05.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034265', '2881205411123', 'Costea Doriana', TO_DATE('09.05.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034266', '6020714129875', 'Antonescu Patricia', TO_DATE('13.05.2026', 'DD.MM.YYYY'));
INSERT INTO mComanda (cID, cnp, nume, data) VALUES ('C034267', '2980930246548', 'Popa Mirela', TO_DATE('18.05.2026', 'DD.MM.YYYY'));
```

```sql
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034260', 1, 'SN-OLED65', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034260', 2, 'SG-HG50U', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034260', 3, 'LN-20XW', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034261', 1, 'LN-55U8', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034261', 2, 'AP-IP17P', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034261', 3, 'SN-WH5', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034262', 1, 'AP-IP14P', 3);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034262', 2, 'LN-20XW', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034262', 3, 'SN-SB400', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034263', 1, 'SN-OLED65', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034263', 2, 'SG-HG50U', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034263', 3, 'LN-L790', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034264', 1, 'AP-MQ24', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034264', 2, 'SN-SB400', 3);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034264', 3, 'LN-55U8', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034265', 1, 'SG-S928B', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034265', 2, 'SN-WH5', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034265', 3, 'AP-IP17P', 1);

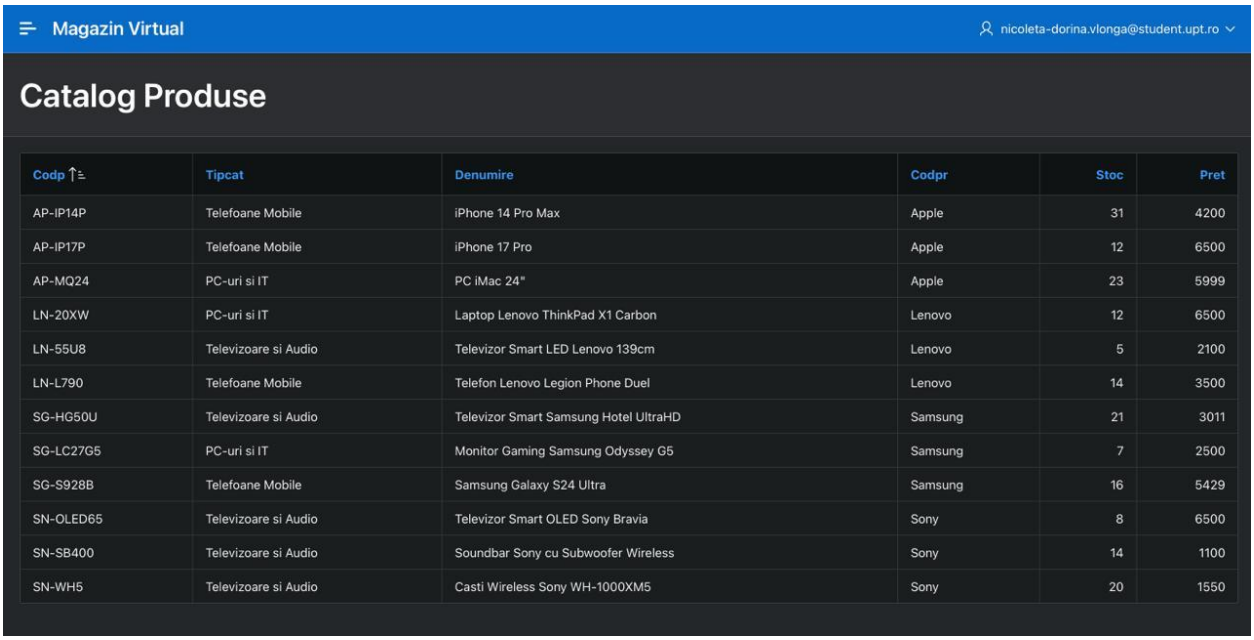
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034266', 1, 'SG-S928B', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034266', 2, 'AP-MQ24', 4);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034266', 3, 'SG-LC27G5', 1);

INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034267', 1, 'AP-IP14P', 2);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034267', 2, 'SG-LC27G5', 1);
INSERT INTO mListaCom (cID, poz, codP, buc) VALUES ('C034267', 3, 'LN-L790', 1);

COMMIT;
```

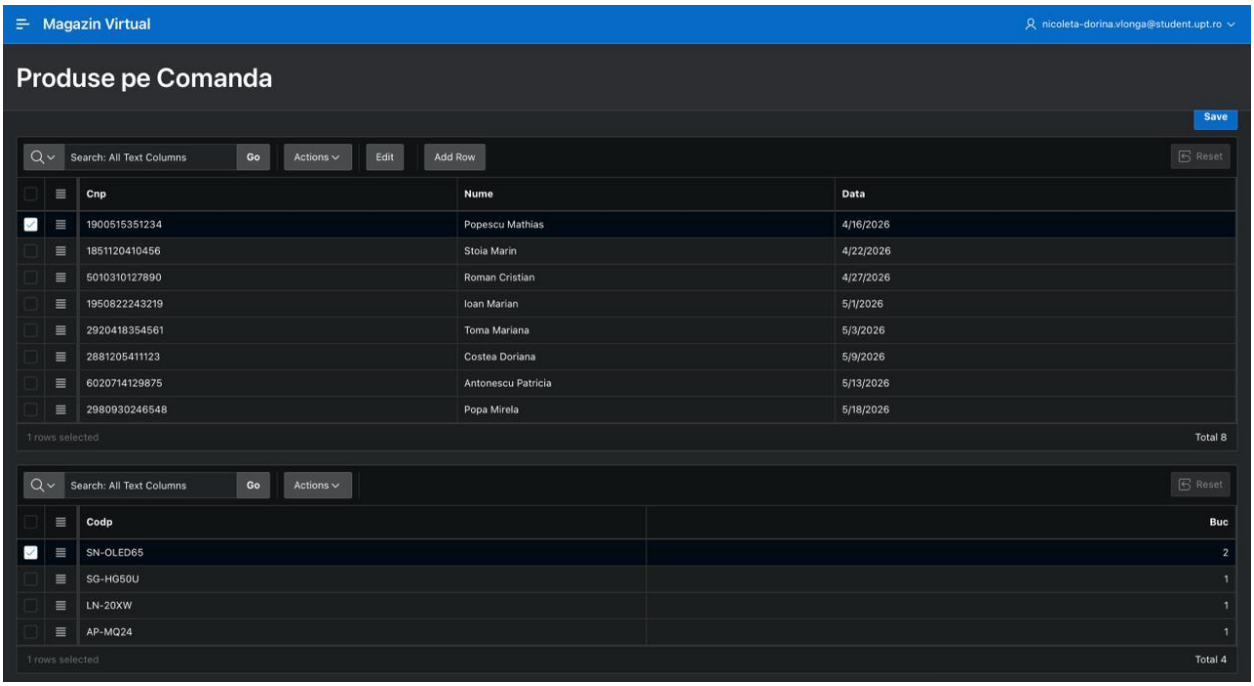
Student A:

6.3)

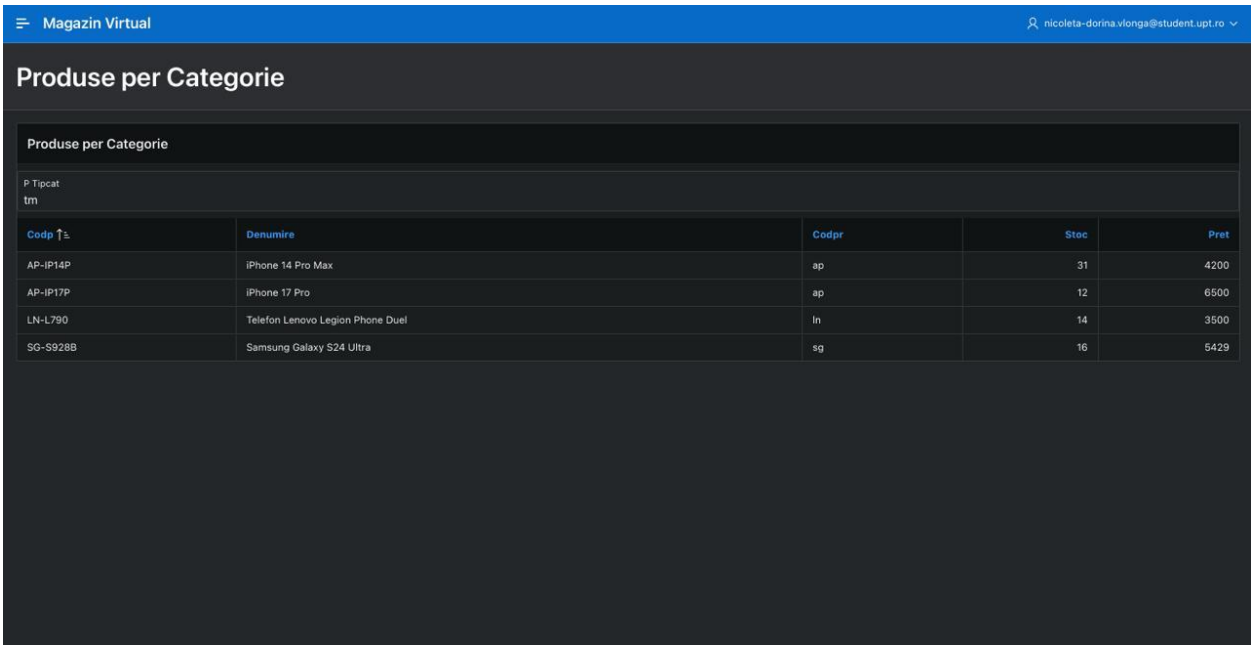


Codp	Tipcat	Denumire	Codpr	Stoc	Pret
AP-IP14P	Telefoane Mobile	iPhone 14 Pro Max	Apple	31	4200
AP-IP17P	Telefoane Mobile	iPhone 17 Pro	Apple	12	6500
AP-MQ24	PC-uri si IT	PC iMac 24"	Apple	23	5999
LN-20XW	PC-uri si IT	Laptop Lenovo ThinkPad X1 Carbon	Lenovo	12	6500
LN-55U8	Televizoare si Audio	Televizor Smart LED Lenovo 139cm	Lenovo	5	2100
LN-L790	Telefoane Mobile	Telefon Lenovo Legion Phone Duel	Lenovo	14	3500
SG-HG50U	Televizoare si Audio	Televizor Smart Samsung Hotel UltraHD	Samsung	21	3011
SG-LC27G5	PC-uri si IT	Monitor Gaming Samsung Odyssey G5	Samsung	7	2500
SG-S928B	Telefoane Mobile	Samsung Galaxy S24 Ultra	Samsung	16	5429
SN-OLED65	Televizoare si Audio	Televizor Smart OLED Sony Bravia	Sony	8	6500
SN-SB400	Televizoare si Audio	Soundbar Sony cu Subwoofer Wireless	Sony	14	1100
SN-WH5	Televizoare si Audio	Casti Wireless Sony WH-1000XM5	Sony	20	1550

6.4)

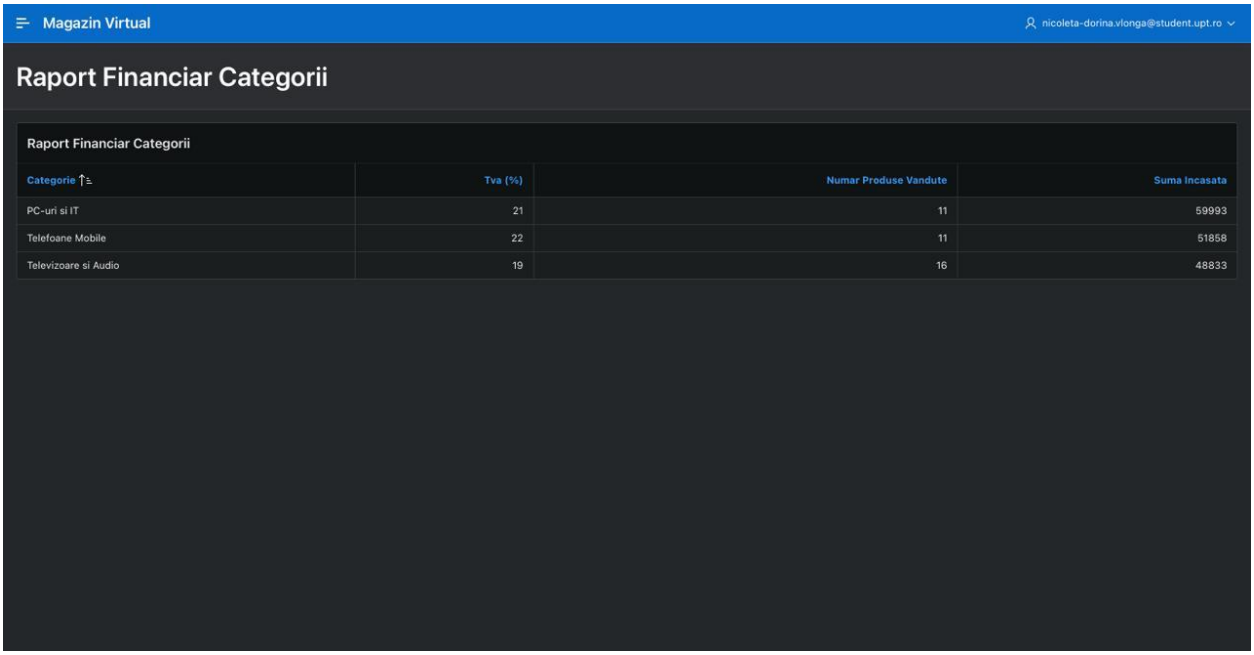


6.5)




```
1    SELECT codP, denumire, codPr, stoc, pret
2    FROM mProdus
3    WHERE tipCat=:P_TIPCAT
```

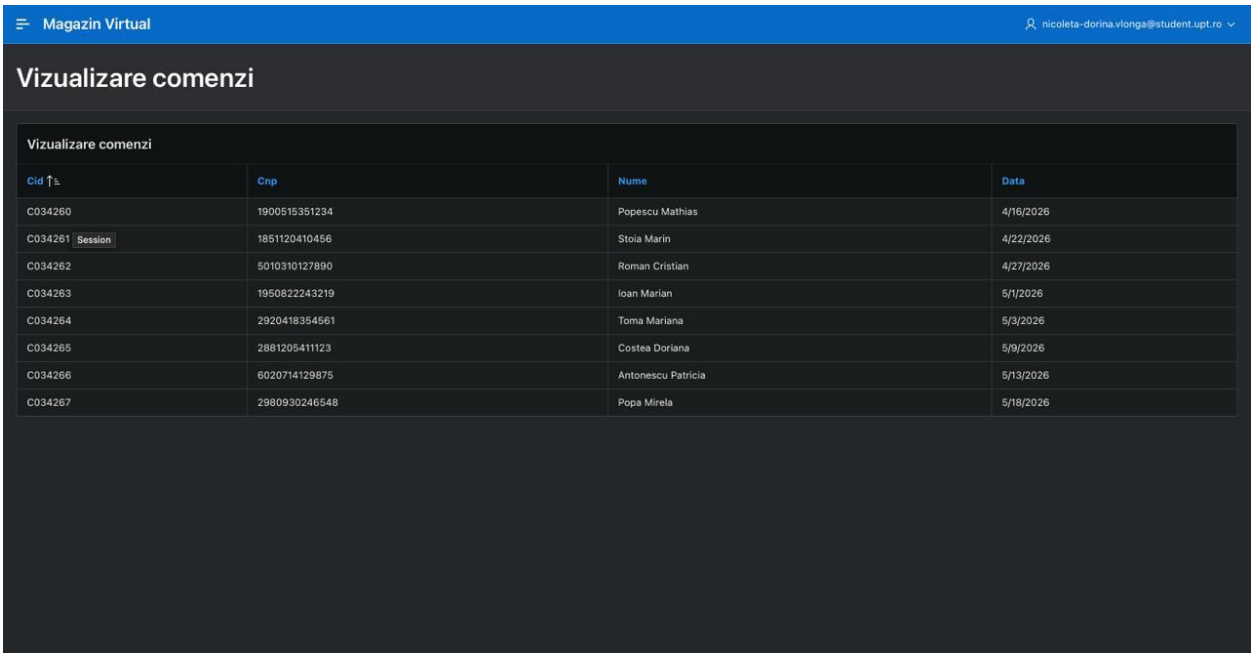
6.6)



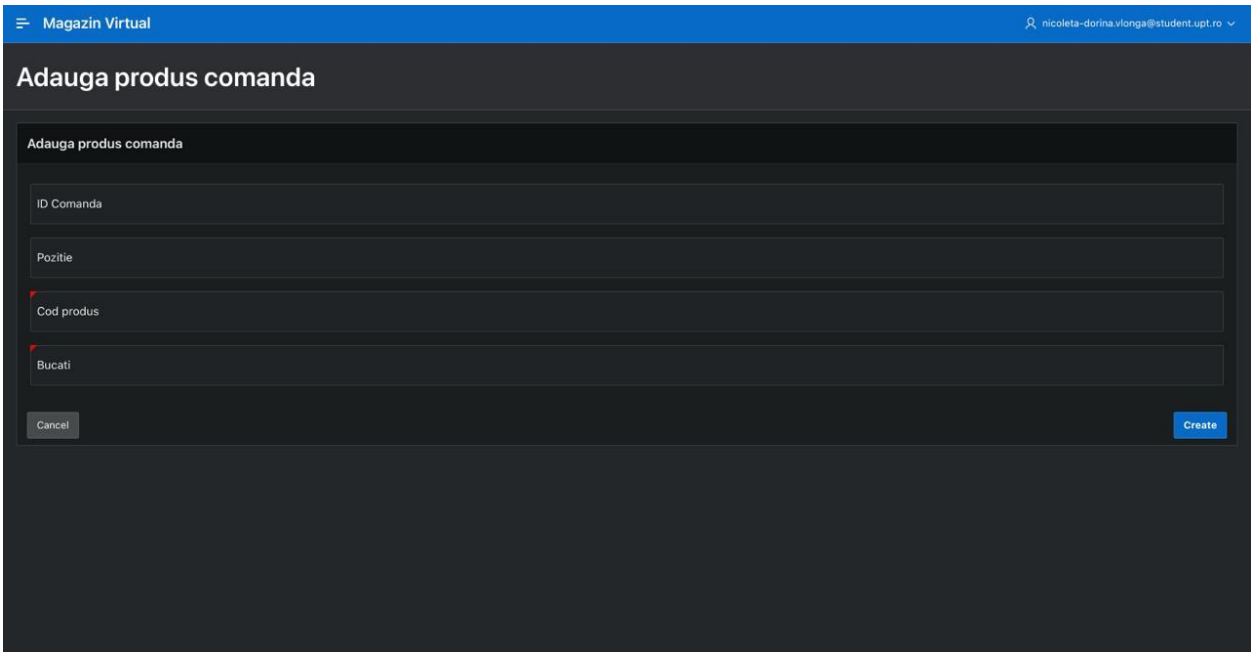
```sql
SELECT c.categorie AS "Categorie",
       c.tva AS "TVA (%)",
       SUM(lc.buc) AS "Numar Produse Vandute",
       SUM(lc.buc * p.pret) AS "Suma Incasata"
FROM mCategorie c
JOIN mProdus p ON c.tipCat=p.tipCat
JOIN mListaCom lc ON p.codP=lc.codP
GROUP BY c.categorie, c.tva
```

Student B:

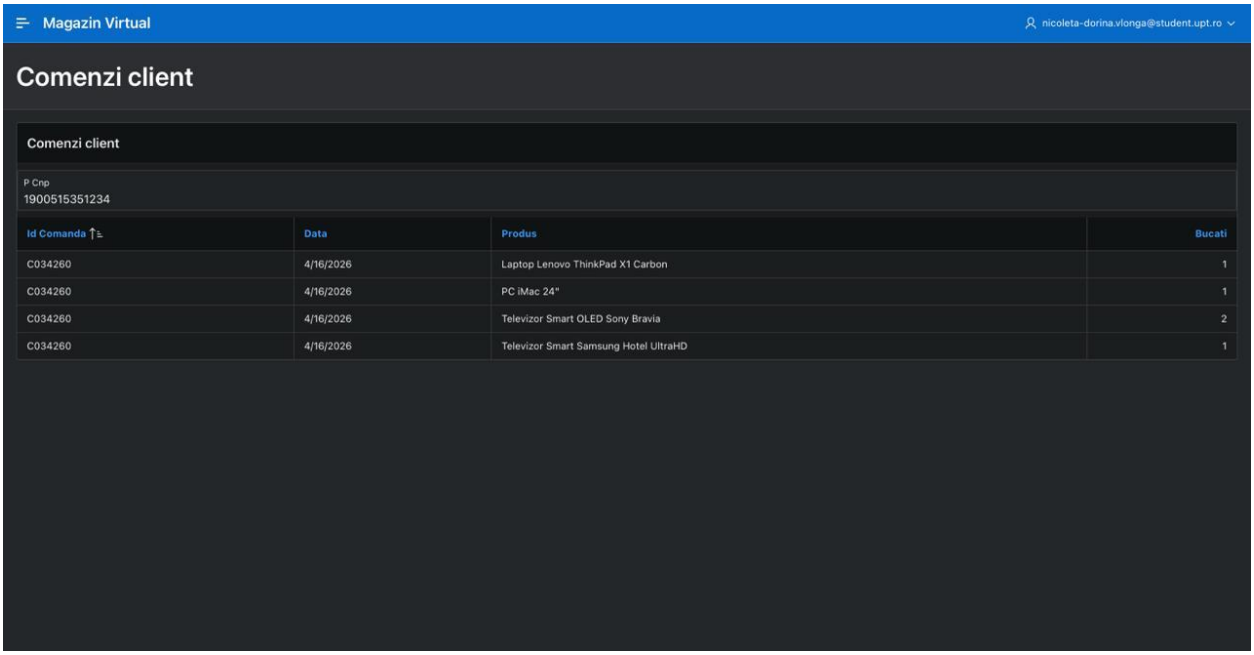
6.3)



6.4)

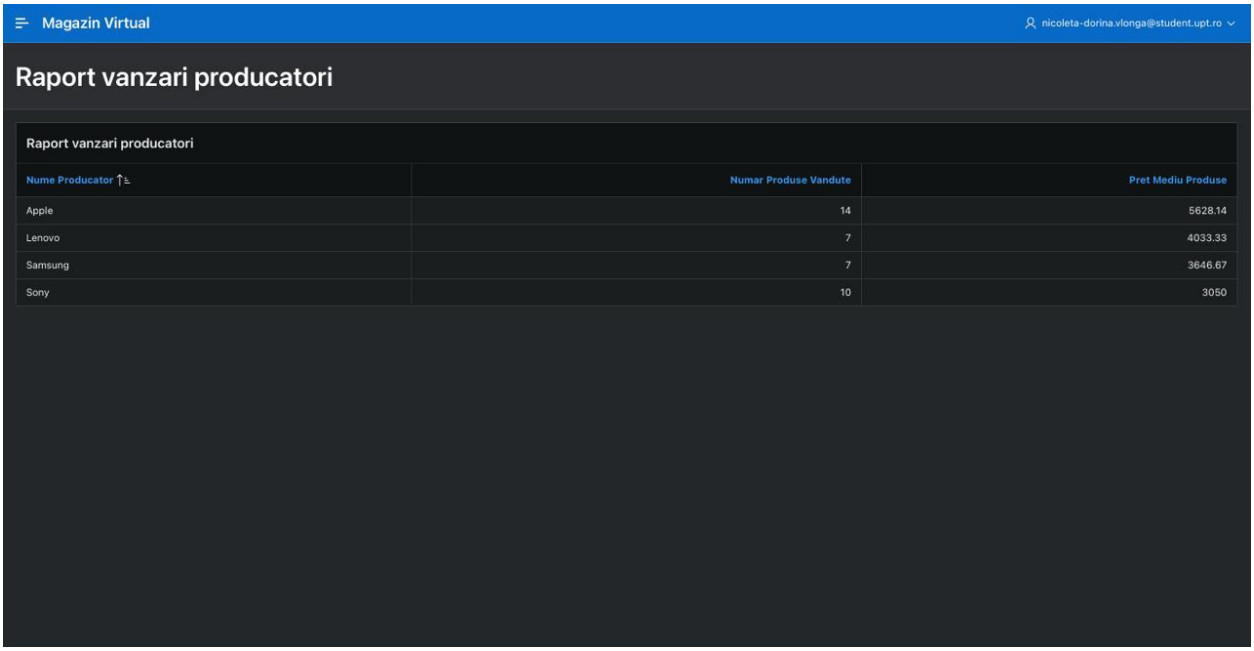

Magazin Virtual
nicoleta-dorina.vlonga@student.upt.ro
Adauga produs comanda
Adauga produs comanda
ID Comanda
Pozitie
Cod produs
Bucati
Cancel
Create

6.5)


Magazin Virtual
nicoleta-dorina.vlonga@student.upt.ro
Comenzi client
Comenzi client
P Cnp
1900515351234
Id Comanda
Data
Produs
Bucati
C034260 4/16/2026 Laptop Lenovo ThinkPad X1 Carbon 1
C034260 4/16/2026 PC iMac 24" 1
C034260 4/16/2026 Televizor Smart OLED Sony Bravia 2
C034260 4/16/2026 Televizor Smart Samsung Hotel UltraHD 1

```sql
SELECT c.cID AS "ID Comanda",
       c.data AS "Data",
       p.denumire AS "Produs",
       lc.buc AS "Bucati"
FROM mComanda c
JOIN mListaCom lc ON c.cID = lc.cID
JOIN mProdus p ON lc.codP = p.codP
WHERE c.cnp = :P_CNP
```

6.6)




```sql
SELECT pr.nume AS "Nume Producator",
       SUM(lc.buc) AS "Numar Produse Vandute",
       ROUND(AVG(p.pret), 2) AS "Pret Mediu Produse"
FROM mProducator pr
JOIN mProdus p ON pr.codPr = p.codPr
JOIN mListaCom lc ON p.codP = lc.codP
GROUP BY pr.nume
```